



// FINANCE & HEALTH HUD

MRLN

THE COMPLETE GUIDE

Offline · **Private** · **30-day money-back guarantee** · one self-contained
file

<https://sagemrln.github.io/my-first-repo/>

◆ 1. What MRLN is (and why it exists)

MRLN is a personal **command center for your money and your body** — one cyberpunk dashboard that turns your real numbers into a plan you can actually follow.

It was built around four hard rules:

PRINCIPLE	WHAT IT MEANS FOR YOU
Paid — so you are not the product	MRLN is a paid app: you buy an access key. There is no account to create, no advertising, and nothing about you is sold. The app itself makes no paid calls to anything, so there is never a per-question or per-use fee on top. Every purchase is covered by a 30-day money-back guarantee — if MRLN isn't for you, email us within 30 days of purchase and we'll refund you (full terms in the Refund Policy on the legal page).
Offline	Everything runs in your browser. Once loaded it works with the plane on airplane mode.
Private	Your data never leaves your device . Nothing is uploaded, tracked, or sold.
Single file	The whole app is one self-contained HTML file — nothing to install, easy to back up.

The purpose: most finance and fitness apps are free because your data is the product — sold, or used to target ads. MRLN is paid instead, so your data never has to be the payment. It gives you the math, the projections, and the honest verdicts — and keeps all of it on your phone.

◆ 2. Getting in — the access key

MRLN opens **locked**. You unlock it once with an **access key** (a code that looks like `MRLN-...`).

- The key is **cryptographically signed** — it can't be faked or guessed.
- It's **time-limited** (a tester key might last a week; a personal key, years). When it expires, the app re-locks — even mid-session — and asks for a new one.
- Once entered, it's **remembered on that device**, so you don't retype it every time.
- Keys are tied to access, **not** to your data. Renewing a key never touches your numbers.
- **Your data is never held hostage**. If your key lapses, the app locks — but a 📦 **Export my data** button sits right there on the lock screen, so you can always take your numbers out, key or no key.

Three ways a key can be scoped:

- **File-bound** — works only on one specific file/link (its "File ID", e.g. `#F-AB12CD`).
- **Universal** — works on any link or installed copy (used for testers and the owner).

On first unlock, you'll accept the legal terms and privacy notice. This is recorded on your device and appears only once.

◆ 3. Languages & currencies

Tap the ⚙️ **gear icon** (top right) and open **Settings** → **Locale** to change language, currency, and country. The **locale controls** (🌐 Language · Currency · Country) live in that group.

Languages (8): English, Español, Dansk, Deutsch, Svenska, Norsk, Magyar, **Français**.

Switching translates **the entire interface** instantly — tabs, headings, buttons, help text, the setup wizard, everything. A built-in **language watchdog** runs constantly: it re-checks every screen — including text that only appears in certain situations (your tier verdict, tax breakdown, calendar, logs) — and keeps it all in your chosen language, with nothing left in English.

Currencies (13): USD, EUR, GBP, DKK, SEK, NOK, CHF, CAD, AUD, JPY, HUF, **SGD, KRW**.

Currency is a **real conversion**, not just a relabel. Switch from kr to € and **8,000 kr** becomes **~1,070 €** at the bundled offline exchange rate. Every stored figure — income, expenses, loan, savings — converts together, and switching back returns the original amount. (Rates are an offline snapshot; no server needed.)

◆ 4. First-time setup (the wizard)

On first unlock, a short wizard sets you up in under a minute:

1. **Welcome** — language, currency, country (or tap *"show me a filled example"*).
2. **Life stage** — a quick question: are you *Studying*, *Working*, or *Managing* others? (No age required — COPPA-safe for all ages.) Used to suggest which app pages matter most to you.
3. **About you** — name (optional birthday/ethnicity*, phone & main device).
4. **Your income** — typical monthly take-home, plus optional slow/good months, debt payment, savings target.
5. **Your expenses** — quick-add chips or type your own.
6. **Sports & hobbies** — added to your calendar as weekly repeats.
7. **Getting around** — transport costs; mark anything *financed* and it drops off your budget automatically once it's paid off.
8. **Prioritize pages** — tap app pages in the order *you* want them (most important first). Your picks are saved with your file. Tap **Skip** to move on.
9. **Install** — tailored guide to add MRLN to your home screen (iPhone / Android / PC).
10. **Review** — your starting picture.

* Ethnicity is used **only** to set accurate WHO health thresholds (BMI/body-fat). It never changes calorie targets.

You can re-run setup anytime with ⚙️ **Setup**, or skip it and fill things in yourself.

◆ 5. The Money side

Overview — *Mission Briefing*

Your headline: left-over money in a typical month, comfort margin, and worst-case month. The single glance that tells you if the plan holds.

Income

Your monthly take-home is tracked three ways: **low**, **typical** (baseline for every simulator), and **good** months.

Two ways to set your income:

1. **Manual statement** — *"I earn 18000 a month"* via  **Assistant MRLN** sets your guess across all three tiers (you can adjust low/good separately in the Income card).
2. **Monthly Income Log** — Log each actual month's take-home as it arrives (tap the Income card → **Log a month**). After **3+ months logged**, the app derives typical = **median** of your last-12 months; low = minimum; high = maximum. This live history replaces your guess automatically. If you enter a new manual statement, it overrides history until you log the next month — then history takes back control.

→ A single glance at the **Overview** tab shows your plan's margin: typical month vs. your expenses.

Tax Helper (inside Income)

An offline **estimate** of your income tax, adapting to your country/region:

- What you'll pay, what to expect back.
- **Denmark:** full model — AM-bidrag, personfradrag, beskæftigelsesfradrag, bund/kommune/kirke/topskat — plus **forskudsopgørelse** suggestions (trækprocent + monthly fradrag) and an **over/under-estimate simulator** (set your expected annual vs. actual → refund or owe).
- Now covers 12 countries with local tax models: **Denmark (DK)**, **United States (US)** by state, **United Kingdom (GB)**, **Germany (DE)**, **Spain (ES)**, **Sweden (SE)**, **Norway (NO)**, **France (FR)**, **Italy (IT)**, **Singapore (SG)**, **Japan (JP)**, **South Korea (KR)** — each with an unofficial estimate using local brackets, employee social contributions, and surtaxes. Adjust any rate in **Advanced** to refine the estimate for your region.
- Generic fallback (**XX**) for all other countries — set your flat tax + social rate.
- Always unofficial — verify with your tax authority. Numbers stay on your device.

Budget — costs at a glance + full edit

A unified view of every recurring cost in one place. Budget opens in a clean read-view; tap Edit to change prices, delete items, or add new costs.

- **Budget Summary** — your total monthly and yearly costs, grouped by category with visual share bars (biggest costs visible at a glance). "Varies" items are left out of the total.

- **Add a Cost** — add any new recurring expense to any category. Your live numbers update everywhere (grand total, cash-flow simulator, savings projection all recompute instantly).
- **Your Costs** — every recurring cost, grouped by category. In Edit mode, you can change a price, billing frequency (monthly / quarterly / yearly), delete **any** item (base or added), and add/remove whole categories. All live numbers recompute as you change them.

Loan & Debt

Track a loan payment, amount, and (optional) interest rate; toggle whether it counts in your monthly math.

Cash Flow — *Cash-Flow Simulator*

Drag the income slider to see any month play out: income – costs – loan – savings = what's left. The range auto-fits your real income. Below it, an **8-year savings-growth projection** charts your pooled savings across low/typical/good scenarios with 1/3/8-year milestones.

Savings — *rules + boxes*

The "**Savings**" tab opens with the short list of **wealth-building rules** (pay yourself first, build a buffer, make it earn, clear pricey debt first, bank your good months) — the habits that do almost all the work.

Below the rules are your **Savings Boxes** — Lunar-style pots. Make a box for each thing you're saving for (a holiday, a new phone, an emergency cushion). Tell it what it **holds now**, what you **add** (monthly / quarterly / yearly, or top up by hand any time), and an optional **interest rate** and **target**. A slider projects every box **X years ahead** — what it becomes, what you put in, and exactly **how much the interest earns you** — with a progress bar and an honest "reach it in ~N years." All boxes total up, and every figure converts with your currency.

Your Life Tier

An honest, $S \rightarrow F$ **rating of your money life**, built **only from your own numbers**, and it moves as those numbers move. The **GRADE** pill at the top of the screen shows the same result at a glance, and **Stats & Grades** shows the factor-by-factor breakdown behind it — slow-month resilience, savings rate, fixed costs against income, free cash each month, and income source.

If you've filled in your body stats, your body grade is blended in and the card shows both parts (💰 Finance and 🏋️ Body) under the headline. Turn the body toggle off in **Gym** and the tier goes back to money only.

Your Checklist

Your own to-do list, on the same tab. **MRLN does not write it for you**. A new checklist is empty and says "*No items yet — add your first below.*" — everything on it is something you put there.

- **Add** — type in the box at the bottom and tap + (or press Enter). Up to 200 characters an item, and as many items as you want.
- **Tick** — tap the box beside an item to mark it done. Tap it again to undo.
- **Delete** — tap ✕ on the item.

It saves with the rest of your data, on your device only, and it comes with you when you move your data to another phone.

Coming from an older version? MRLN used to print its own action plan on this tab. That plan has been removed, and the tab tells you so: *"The built-in plan has been replaced by your own checklist."* None of your money numbers changed — only the built-in list is gone, and the space is yours now.

◆ 6. The Health side

Gym Plan

- **Body Stats** — sex, age, height, weight, activity, body-fat. Feeds everything below.
- **Body Grade & Shape** — an S → F physique grade, body-fat/muscle read, and a visual silhouette.
- **Body Measurements** — track measurements over time.
- **Calorie Simulator** — your BMR/TDEE and a slider for cut / maintain / bulk, with honest verdicts ("aggressive deficit", "lean surplus", etc.).
- **Meal Ideas** — a **155-meal** library, matched to your body and your goal. Pick the **physique you want** (balanced / lean / athletic / muscular) and your **training focus** (general / strength / calisthenics / hybrid); MRLN works out your per-meal calorie and protein target from your stats and shows you **how much** of each meal to eat — *"2× serving"* — not just what to eat. Type ingredients or cuisines you **like** and anything to **avoid**; matches are pinned to the top of every page and a dislike is never suggested. Typo-tolerant ("avoid chesse" still works). Meals are tagged ★ **your taste**, ☒ **HEALTHIEST FOR YOU** (the best fit for your body among the meals that suit your goal), and **OFF YOUR GOAL** when a meal you asked for doesn't match the goal you set. **Tap any meal** for the full window: ingredients, your scaled calories and macros, why it fits you, and a **recipe-video link** (that link is the one thing that goes off-device, and only if you tap it — it opens YouTube with the meal's name as the search, and sends nothing else). Press **Suggest** again to rotate through the rest.
- **Personal Records (PRs)** — log strength lifts (weight/reps → auto-calculates 1RM) and cardio (distance/time → tracks pace). Both group by exercise and show your best. **Workouts / Training Split** — your weekly plan.

Food Log

- **Log a meal by typing** ("2 eggs and rice") — an offline estimator reads quantities/units and returns calories + macros — and/or attach a **photo**.
- **Today** total, **History**, delete — your daily food diary, fully offline.

◆ 7. Organize & track

Calendar

Add dated events and notes; one-off, weekly, monthly, or yearly repeats (birthdays 🎂 included). Hobbies and transport from setup land here automatically. **Reminders** 🔔 included.

Stats & Grades

A roll-up of your grades and key stats across money and body.

Notebook

A built-in notes app (like iPhone Notes) — title + body, autosaved, private.

◆ 8. Media Log — rate films, shows, and games

MRLN's Media Log is a personal **rating journal for films, series, shows, and games** — not a social network, just a ranked list of what you've watched/played and what you think.

How it works:

- **Add** — type a title and choose its type: **Film**, **Show**, or **Game**. Add an optional **Genre** (e.g. "Sci-Fi", "Drama"). It goes to your "To-watch" queue.
- **Filter by type** — tabs at the top let you view **All**, just **Films**, just **Shows**, or just **Games**. Your filter stays while you browse.
- **IMDb link** — each entry has a ↗ **IMDb** button that opens IMDb in a new browser tab with the title pre-filled. It sends only the title to IMDb — never your MRLN data (no finance ID, and rel=noopener noreferrer strips the referrer so IMDb can't even see which app you came from).
- **Rate** — open an entry and rate it 1–10 in **0.1-step increments** (e.g. 8.4, 7.6, 9.0). The rating is locked to [1, 10] with one decimal place.
- **Comment** — add a private note (spoilers welcome — it's yours alone). Comments stay on your device.
- **Auto-ranked** — your watched entries sort **by rating descending** — highest-rated at the top, ties broken alphabetically. To-watch entries stay in a separate list.
- **Calibration prompt** — when you rate something, MRLN shows you up to 3 similar-rated entries (within ± 0.2 of your new rating) and asks *"does this belong here?"* — a quick honesty check. The prompt vanishes once you confirm (or you can ignore it).
- **Search** — search by title or comment; matches highlight in both lists.
- **Share** — tap the share button to copy your ranked list (top 20) as plain text — paste it anywhere. Formatted as a numbered list: `1. Title 9.4`, etc.

Data: Rating, comment, type, genre, date added, date rated, and title — all stored locally. **Your Media Log data stays on your device — it never syncs or uploads on its own** (the ↗ IMDb button is the one exception, and only when you tap it: it sends just the title, nothing else). Your

Media Log **does** come with you in the quick clipboard move — titles, types, genres, ratings, comments and dates all travel with the rest of your data (see §10), so your list is the same on your new phone.

◆ 9. Installing the app

Installing is **front and centre**, not buried in a menu:

- **Lock screen** — before you even sign in, short **"Add to Home Screen"** guides for iPhone and Android sit beside the unlock card.
- **"Get the app" banner** — a bright banner across the top of the dashboard. Tap **Install** and it either installs in **one tap** (Android / Chrome / Edge) or opens a **step-by-step** card for your exact device (iPhone Safari, Android, or desktop).
- It **removes itself automatically** the moment the app is installed — no clutter once you're set up.

◆ 10. Connect — move data between devices

The hub for moving your data — all **on-device**, nothing is ever uploaded.

📦 Move my data (new phone, or just installed the app)

This is the simple way to carry everything across. The app reads your situation and puts the **right step first** (📦 *bring in* when you're on a fresh install, 📦 *copy off* when you already have your data here), and labels them **STEP 1 / STEP 2** so it's clear which to do first.

- 📦 **Bring my data onto this device** — already tapped **"Copy my data"** on your other phone or browser? Tap "📋 **Bring in my data**" and MRLN finds what you copied and loads it all. *(On iPhone, Safari and the installed app keep separate data, so this is how your numbers get across — your phone may ask permission to paste; that's normal, just allow it.)* Can't paste? Open **"Can't paste? Do it by hand"** to paste the text or pick a saved file — or load a whole saved **master / exported** `.html` in one tap (income, expenses, notes,
- 📦 **Copy my data off this device** — tap **"Copy my data"**, then open MRLN on the other device and tap **"Bring in my data."** Need another route? **"Other ways to send"** shows your data **code** to copy, or lets you **download it as a file**.

Importing **replaces** what's on the device you bring it onto, and the app ****asks you to confirm**** before anything changes.

Note on photos: Meal photos are stored separately on each device (not in the transfer code), so they don't ride along when you move your data — only the meal metadata (date, calories, description) transfers. When you bring your data to a new phone, your meals come across, but

photos stay on the old device. You can re-photograph the meals on the new phone, or export/import meals individually if you need the photos (Apple Health, CSV).

Other Connect tools

- 📁 **Import from another app** — bank/card **CSV** → income & expenses; **.ics** → calendar; Apple Health/CSV → weight; MyFitnessPal CSV → food log.
- 📁 **Export to other apps** — your calendar as **.ics**, expenses/food as **CSV**, or the native **Share** sheet.
- 📱 **QR transfer** — move your link/data by scanning a QR code (generated fully offline).
- ⚖️ **Bluetooth scale** (in Gym) — read your weight straight from a compatible scale.

◆ 11. 🗨 Assistant MRLN — offline Q&A + plain-language commands

Your personal finance & health assistant. Tap 🗨 **Assistant MRLN** to ask questions about **any feature** (how does savings work? what is the food log?) or **change your data** with plain-language commands (one per line). Everything runs **offline, in your browser** — no server, no per-question fee, and nothing changes until you confirm.

Questions: Ask questions in **English**. The assistant answers from a complete knowledge base covering every tab and feature. You can also ask about **your own numbers** and it computes the answer directly from your data, all locally and offline:

- **Income** — *"what is my typical income?", "income low month?"*
- **Spending by category** — *"where does my money go?", "what are my biggest costs?"*
- **Savings** — *"how much have I saved total?", "savings in Emergency Box?", "how close to my target?"*
- **Weight** — *"what do I weigh?", "have I lost weight?"*
- **Affordability & savings time** — *"can I afford 5000?", "how long to save 50000?"*

Commands: Update your numbers using natural language — type the way you'd naturally say it. It understands commands in the same 8 languages (e.g. *"jeg sparer 400 om måneden"*, *"min alder er 41"*, *"mon revenu est 25000"*). Your item, employer and note names are always kept exactly as you typed them. It understands:

- **Income** — *"income is now 2600", "good month income is 22000"*
- **Costs** — *"add Gym 29/mo to Other", "rent is now 1200", "cancel Netflix"*
- **Loan / savings** — *"loan payment is 1500", "save 400 per month"*
- **Categories** — *"add category Pets 🐾"*
- **Profile** — *"name is Alex", "work is now Acme Ltd", "I'm 24"*
- **Body** — *"I weigh 82kg", "height is 180"*
- **Calendar** — *"note 2026-07-15 Mum's birthday"*

- Workouts — *"workout Monday: push day"*

◆ 12. Daily Habits — engagement & motivation

A handful of small mechanics keep the app worth opening daily:

Streak Counter — Track consecutive days you've logged income or expenses. The header shows a **STREAK N** pill with your count — green when you logged something today, amber when it's been a day or two. Breaking a streak resets you to 1, so dailies become a ritual. It's the oldest retention mechanic in apps like Duolingo and Habitica — and it works.

☀ **Morning Briefing** — Once per day on app load, you see a time-aware greeting by name. *"Good morning, Ingrid 🙌 — 7 days in a row · on track to keep €2,140 this month"* The greeting changes by time of day (Good morning/afternoon/evening); combines your streak count and your **monthly surplus** (the amount you're on track to have left over after expenses and loan payments). **Offline-only, local-only** — never leaves your device. A daily reminder of your progress without hype.

Apply Feedback — When you apply a command, a short toast pops up with the count of what changed — *"✓ 3 applied"*. It's instant proof that the app heard you and that the numbers on screen are the new ones.

🔊 **Satisfying UI Sounds** — Synthesized audio chimes (Web Audio, no files, fully offline) play on key actions. The core rule: **sounds voice outcomes, not inputs** — you hear confirmation when something works or changes, but not on every keystroke or slider tick. This keeps the app feeling responsive without becoming grating.

- **Always sound:** tab navigation (soft tick), Apply success (C–E–G chime), save/export (two-note), streak milestones (celebration fanfare).
- **Never sound:** keystrokes, page loads, or scrolls.
- **Slider drag sound:** when you drag any simulator slider (cash flow, calorie, savings projection), it plays a pitch-tracked tone that rises with the value — giving real-time audio feedback as you move the slider, without noise or distraction. Fully throttled so it stays musical, not noisy.
- **Optional (context-aware):** modal open (subtle entrance), delete (down-whoosh if you've just deleted something), income-up alerts, error feedback — these respect your usage pattern and throttle so rapid actions don't spam.

Sounds are **soft by design** (~0.09–0.12 gain) and **respect reduced-motion** mode: if your OS is set to prefer reduced motion, sounds default OFF (toggle back ON anytime with 🔊/🔕). Even with sound ON, the visual toast (✓ or ✨) always confirms every action — sound is the *bonus*, never the only signal.

These mechanics are **per-device, never synced** (they reset if you import to a new phone) and are **stripped from any file you share** — so a customer file or a test upload stays clean, and the habit counters don't follow your export.

◆ The Dashboard — navigation & home screen

The dashboard organizes all your data into **four main sections** plus a gear menu:

Navigation (a bottom dock on phones; a left sidebar on wide screens):

- **Home** — overview and stats
- **Money** — Income, Budget, Loan, Cash Flow, Savings, Checklist
- **Health** — Gym Plan and Food Log (body stats are a card inside Gym Plan)
- **Life** — calendar, notebook, media log
-  **Gear** (top right, header icon) — Settings, Connect (move your data), and Change Log
-  **Assistant** — dock cell to open the offline Q&A assistant

Tap any group to open its sections; tap again to collapse.

Home Screen — at a glance

The **Home** tab (default first tab) gives you your financial picture in one screen:

1. **Hero number: "Left this month"** — What remains after every bill, debt payment, and before savings, in a typical month. The number you watch.
1. **Income scale:** A row labeled **"If your income is..."** shows three reference points — Low Month / Typical / High Month — derived from your actual income history (or your guess if you're just starting). This isn't a choice; it's the range your app uses internally for all simulators and projections. At a glance, you see how far apart your low/typical/good months really are.
1. **Income history** — A mini-chart that appears once you have logged real months; it shows up to your last 12 logged months, with the first and last month labelled and the latest value shown above its bar.
1. **Four KPI cards** — Four core metrics, each one tap-able to go deeper:
 - **Calories** — today's total, with your daily kcal target in the sub-line
 - **Protein** — today's total (grams), with carbs and fat breakdown in the sub-line
 - **Weight** — current weight and change since your previous logged entry
 - **Workouts** — count of workouts in your plan, with a sub showing "plan · no dates yet" (plan entries have no scheduled dates)

Each card is a button; tap it to go to the detailed page (Gym for workouts, Food for calories/protein, Gym weight log for weight history).

Weight Log — tracking body changes

Under the Gym tab, the **dated weight log** records your weight history with dated entries. You can log weight three ways:

- **Weight field** — type your weight into the Weight field in your body stats (Gym tab); each entry is auto-dated when logged.
- **Bluetooth scale** — connect to a compatible scale and it logs automatically.

- **CSV import** — upload a CSV file (date · weight columns); the app reads the most recent row and logs it as one dated entry.

The log shows all your entries; the Home Weight card displays your current weight and the change since your previous logged entry.

Settings Panel

Tap the  **gear icon** (top right) to open the Settings panel. It's a full-screen sheet on phones, or a 560px card on desktop (the dock stays visible beside it). The panel groups controls into four sections:

1. **Locale** — Language (change which of the 8 languages the whole interface reads in) · Currency (pick your currency and see every figure in that denomination) · Country (set your region for tax math and health thresholds)
2. **Appearance** — Theme (9 themes: Cyberpunk/Amethyst/Glacier/Evergreen/Carmine/Rose/Umber/Ember/Daylight) · Layout (options for how the dashboard fits your screen)
3. **Help** — Tour (guided walkthrough of every page) · Setup (the first-time wizard, run anytime) · Sound (on/off toggle for UI chimes — turn off if you prefer silent)
4. **Data** — Open / Save (the one button to save your data as a file, or load a saved file back in)

Each section shows its heading and controls inline. Changes save instantly. A **Back** button sits top-left to close the panel. The Settings panel is the only way to reach these controls — there is no "Accounts" or "Profiles" area.

◆ 13. Updates — how new features reach you

- **On the hosted link / installed app:** updates are **automatic**. Open it online and the newest version loads itself (network-first service worker); offline still works from cache. Your data is preserved across every update — no migration, no re-login.
- **On a downloaded file:** that file is frozen. To get new features, open the hosted app (or a newer file) and **import your data** across.

Tip: running your master *on the installed app* gives you the best of both — your data + auto-updates.

◆ 14. Privacy & anti-tamper (the security model)

- **Your data is yours.** It lives in your browser's local storage (and embedded in your saved file). It is never uploaded — there's no server to upload to.
- **Unforgeable access.** Keys are signed with a private key that exists **only** in the owner's master file. The public app holds just the public half — enough to verify keys, never to make them.

- **Tamper = poison.** Three independent watchdogs detect a bypassed lock (overlay removed, public key swapped, verifier neutered). If anyone defeats the lock, an integrity token flips and **poisons every number** — finance, body/calories, tax, and food all become `NaN` — and a full-screen warning appears. A broken app is worthless to a thief.
- **Clean shares.** Any blank/customer copy is **hard-stripped** of all personal data and of the owner key, so nothing private can leak into a file you hand out.
- **Accidental selection blocked.** Dashboard numbers and labels can't be accidentally selected or searched — long-pressing the screen won't trigger "copy" or "web search" menus that could expose private figures. Your access key, data code, and all input fields stay fully copyable and typeable.

◆ 15. For the owner — running the show

These tools appear only on **your master file** (where the private key lives):

-  **New / renew key** — mint a key for anyone: pick a name and duration (minutes → years). Optionally bind it to a specific File ID, or leave it universal.
-  **New file + key** — birth a fresh, blank customer file **and** the single key that unlocks it, together. (Never share your master — only the file+key it creates.)

Releasing updates: edit `index.html`, then bump **both** `APP_VER` and the `VERSION` in `sw.js` **every time a published file changes** — that version is what makes an already-installed copy throw away its cached page and fetch the new one. Skipping it is why installed apps used to sit on an old build. Run `node tools/release/green.js` (it fails the release if anything is red) and `node tools/publish/deploy.js --check` before pushing to `gh-pages`; everyone on the hosted link or installed app picks it up on their next online open.

Your master file auto-unlocks on your devices, holds your real data, and can mint keys. Keep it private; back it up; never hand it out.

◆ 16. Quick troubleshooting

SYMPTOM	FIX
"This key expired"	Get/mint a new key. Keys are time-based by design.
"This key belongs to a different file"	You're using a file-bound key on the wrong file. Use the key that came with this file, or a universal key.
Installed app asks for a key again	Installed apps have separate storage from the browser tab. Enter your key once in the installed app; a universal key avoids file-ID mismatches.
Opened the link and it's blank with no key prompt	A bare link (no <code>#F-...</code>) opens a fresh empty app, and/or your browser already had a saved key. Your data lives in your file — import it from Connect.
Can't find where to edit a cost	All costs are in the Budget tab — scroll down to "Your Costs" to edit prices, billing, delete items, or add new categories.
Numbers show as NaN + a red warning	The tamper guard tripped. Use the original, unmodified app with a valid key.

MRLN — your money and your body, on your terms, on your device.